

ePG Introduction to Machine Learning Quiz 1

Solutions

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Solutions

Question 1: Naïve Bayes Log-Odds Derivation

Derive:

$$\log \frac{P(y = 1|x)}{P(y = 0|x)}$$

Answer:

$$\log \frac{P(y = 1|x)}{P(y = 0|x)} = \log \frac{P(y = 1)}{P(y = 0)} + \log \frac{P(x|y = 1)}{P(x|y = 0)}$$

Using conditional independence:

$$\begin{aligned} P(x|y) &= \prod_{j=1}^d P(x_j|y) \\ &= \log \frac{P(y = 1)}{P(y = 0)} + \sum_{j=1}^d \log \frac{P(x_j|y = 1)}{P(x_j|y = 0)} \end{aligned}$$

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Question 2: Bernoulli Naïve Bayes Linearity

Show explicitly that Bernoulli Naïve Bayes leads to a linear classifier.

Answer:

$$P(x_j|y) = \theta_{jy}^{x_j} (1 - \theta_{jy})^{1-x_j}$$

Substitute:

$$\log \frac{P(x_j|y = 1)}{P(x_j|y = 0)} = x_j \log \frac{\theta_{j1}}{\theta_{j0}} + (1 - x_j) \log \frac{1 - \theta_{j1}}{1 - \theta_{j0}}$$

Rearranging:

$$= x_j \cdot \omega_j + c_j$$

Summing:

$$= \epsilon_0 + \sum_j \omega_j x_j$$

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Question 3: Entropy Calculation

Given a node with:

$$P(y = 1) = 0.6, \quad P(y = 0) = 0.4$$

Compute entropy.

Answer:

$$H = - \sum p \log_2 p$$

$$H = -(0.6 \log_2 0.6 + 0.4 \log_2 0.4)$$

$$= -(0.6(-0.737) + 0.4(-1.322)) \approx 0.971$$

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Question 5: Gini Index

Compute Gini index for:

$$P(y = 1) = 0.7, \quad P(y = 0) = 0.3$$

Answer:

$$G = 1 - \sum p^2$$

$$= 1 - (0.7^2 + 0.3^2) = 1 - (0.49 + 0.09) = 0.42$$

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Question 6: Entropy vs Gini

Prove that entropy is maximized at uniform distribution.

Answer:

For binary case:

$$H(p) = -p \log p - (1 - p) \log(1 - p)$$

Differentiate:

$$\frac{dH}{dp} = -\log p + \log(1 - p)$$

Set to zero:

$$\log p = \log(1 - p) \Rightarrow p = 0.5$$

Thus entropy is maximized at uniform distribution.

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Question 7: Decision Tree Split using Entropy and Gini

Consider the following dataset:

Sample	x_1 (Outlook)	x_2 (Windy)	x_3 (Humidity)	y (Play)
1	Sunny	No	High	No
2	Sunny	Yes	High	No
3	Overcast	No	High	Yes
4	Rain	No	High	Yes
5	Rain	Yes	Normal	No
6	Rain	No	Normal	Yes
7	Sunny	No	Normal	Yes
8	Sunny	Yes	Normal	No

- (a) Compute the entropy of the dataset.
- (b) Compute Information Gain for splitting on x_1 (Outlook).
- (c) Compute Gini index for the split on x_2 (Windy).
- (d) Which feature would you choose as the root using entropy? Using Gini?

Answer:

(a) Entropy of dataset

Total: 8 samples Yes = 4, No = 4

$$H = - \left(\frac{4}{8} \log_2 \frac{4}{8} + \frac{4}{8} \log_2 \frac{4}{8} \right) = 1$$

(b) Information Gain for x_1 (Outlook)

Split:

Sunny: 4 samples $\rightarrow$ (Yes=1, No=3)

$$H_{\text{sunny}} = - \left(\frac{1}{4} \log_2 \frac{1}{4} + \frac{3}{4} \log_2 \frac{3}{4} \right) \approx 0.811$$

Overcast: 1 sample $\rightarrow$ pure $\rightarrow H = 0$

Rain: 3 samples $\rightarrow$ (Yes=2, No=1)

$$H_{\text{rain}} \approx 0.918$$

Weighted entropy:

$$\begin{aligned} H_{\text{split}} &= \frac{4}{8}(0.811) + \frac{1}{8}(0) + \frac{3}{8}(0.918) \\ &= 0.4055 + 0 + 0.344 = 0.7495 \end{aligned}$$

Information Gain:

$$IG = 1 - 0.7495 = 0.2505$$

(c) Gini index for x_2 (Windy)

Split:

Windy = Yes: 3 samples $\rightarrow$ (Yes=0, No=3)

$$G_{yes} = 1 - (0^2 + 1^2) = 0$$

Windy = No: 5 samples $\rightarrow$ (Yes=4, No=1)

$$G_{no} = 1 - \left(\frac{4}{5}\right)^2 - \left(\frac{1}{5}\right)^2 = 1 - (0.64 + 0.04) = 0.32$$

Weighted Gini:

$$G = \frac{3}{8}(0) + \frac{5}{8}(0.32) = 0.2$$

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(d) Feature selection

- Entropy criterion $\rightarrow$ choose feature with highest Information Gain $\rightarrow x_1$
- Gini criterion $\rightarrow$ choose feature with lowest Gini $\rightarrow x_2$

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Conclusion:

Different impurity measures can lead to different splits at the root.